Error-Sensitive Alignment Bounds for Affine-Line Decoding: A Lean-Verified Refinement

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Abstract

Goyal and Guruswami derive mutual correlated agreement (MCA) for the affine-line generator from line decodability, and ask for $n+1$ aligned challenges, where n is the block length. We show that $\max(\lfloor \delta n \rfloor + 2,\ 3)$ aligned challenges suffice when $n < |\mathbb{F}|$, where δ is the proximity parameter. The proof replaces an injection of challenges into coordinates by a double count of mismatch coordinates, using that every close challenge misses at most $\lfloor \delta n \rfloor$ coordinates. The same change in the boosting step lowers the main term of the capacity-regime MCA bound for folded Reed–Solomon codes from nt to $(\delta n + 2)t$: $\varepsilon_{\mathrm{mca}}(C, \delta) \le ((\delta n + 2)t + 3t^3)/|\mathbb{F}|$ instead of $(nt + 3t^3)/|\mathbb{F}|$, with $\delta = 1 - \rho - 2/t$. The gain is a constant factor of about $1/\delta$ on the main term. All statements are proved in Lean 4 on top of ArkLib, without `sorry` and with no axioms beyond the standard ones.

1 Introduction

Proximity gaps and correlated agreement are the soundness engine of FRI-type protocols. Mutual correlated agreement (MCA) asks that, for all but few challenges γ, if $u_0 + \gamma u_1$ agrees with a codeword on a set S of at least $(1-\delta)n$ coordinates, then u_0 and u_1 both agree with codewords on the same S.

Goyal and Guruswami [1] introduce *line decodability* (Definition 3.1): whenever many challenges have a decoded codeword close to the line, a positive fraction of them lie on one affine pencil $c_0 + \gamma c_1$. Their Theorem 3.5 turns $(\ell, \delta, a, \ell n + 1)$ curve decodability into MCA error a/q; for lines this asks for $n+1$ aligned challenges. Since line decodability is usually established for a small number of aligned challenges, it is first boosted using a list-size bound (their Theorem 3.6 and Corollary 3.9), and the boost to $n+1$ costs a term nL, giving error $(nL + a)/q$. The same n appears in the folded Reed–Solomon (FRS) bound [1, Corollary 4.10], $\varepsilon_{\mathrm{mca}} \le (nt + 3t^3)/q$, and in later work [2].

Contribution. The threshold $n+1$ comes from an injection: each aligned challenge produces a coordinate where the line agrees with the pencil but (u_0, u_1) does not, and distinct challenges produce distinct coordinates. We observe that each such coordinate also lies in the agreement set of its challenge, which misses at most $e = \lfloor \delta n \rfloor$ coordinates. A double count then bounds the aligned challenges by $e + 1$ (Lemma 3). Consequences:

- MCA error a/q from (δ, a, b) line decodability for every $b \ge \max(e + 2, 3)$, when $n < q$ (Theorem 4).

- The boost can aim at $e+2$ instead of $n+1$, at cost $(e+1)L$ instead of nL (Lemma 5).
- For FRS codes, $\varepsilon_{\mathrm{mca}}(C,\delta) \le ((\delta n+2)t+3t^3)/q$ (Corollary 7).

The counting lemma itself is not new: the same argument appears as `exists_common_affine_set` in the 68.15-bit Proximity Prize submission `016274d1-9b09-4ce1-add3-4c632dbbc283` (Apache-2.0) [5], where it serves an alignment bound for Reed–Solomon codes. We did not find the error-dependent threshold for line decoding, or the resulting FRS bound, in [1, 3, 2]; the argument is elementary and may be known.

All results are formalized in Lean 4 against ArkLib [6] (Section 6).

2 Setting

Let ι be a finite set of $n \ge 1$ coordinates, $\mathbb{F}$ a finite field with q elements, A an $\mathbb{F}$-module (the alphabet; $A=\mathbb{F}$ for Reed–Solomon codes and $A=\mathbb{F}^s$ for folded codes), and $C \subseteq A^\iota$ an $\mathbb{F}$-linear code. Fix $\delta \in [0,1)$ and put $e = \lfloor \delta n \rfloor$. A word x is *δ-close* to y if they agree on at least $n-e$ coordinates. For $S \subseteq \iota$ write $C|_S$ for the restriction of C to S.

Definition 1 (bad challenge)**.** For $u=(u_0,u_1) \in (A^\iota)^2$, a challenge $\gamma \in \mathbb{F}$ is *bad* if there is $S \subseteq \iota$ with $|S| \ge (1-\delta)n$ such that $(u_0+\gamma u_1)|_S \in C|_S$ but $u_0|_S \notin C|_S$ or $u_1|_S \notin C|_S$. The MCA error $\varepsilon_{\mathrm{mca}}(C,\delta)$ is the maximum over u of $\Pr_\gamma[\gamma \text{ bad}]$, with γ uniform in $\mathbb{F}$.

Definition 2 (line decodability [1, Def. 3.1])**.** C is (δ,a,b) line decodable if for all $f_0,f_1 \in A^\iota$ and all $V : \mathbb{F} \to C$ with $\Pr_\gamma[V(\gamma)\ \delta\text{-close to } f_0+\gamma f_1] \ge a/q$ there are $c_0,c_1 \in C$ with $\Pr_\gamma[V(\gamma)\ \delta\text{-close to } f_0+\gamma f_1 \text{ and } V(\gamma)=c_0+\gamma c_1] \ge b/q$.

3 The counting lemma

Lemma 3. *Let $u_0,u_1,c_0,c_1 \in A^\iota$, $E \subseteq \mathbb{F}$ and $e \in \mathbb{N}$. Suppose that for every $\gamma \in E$ there is $G_\gamma \subseteq \iota$ with*

(A1) $|G_\gamma| \ge n-e$;

(A2) $u_0(i)+\gamma u_1(i) = c_0(i)+\gamma c_1(i)$ for all $i \in G_\gamma$;

(A3) some $i \in G_\gamma$ has $(u_0(i),u_1(i)) \ne (c_0(i),c_1(i))$.

Then $|E| \le e+1$.

Proof. Let $M = \{i : (u_0(i),u_1(i)) \ne (c_0(i),c_1(i))\}$ and $d_k = u_k - c_k$. If $i \in G_\gamma \cap G_\beta \cap M$ with $\gamma \ne \beta$, then by (A2) $(\gamma-\beta)\,d_1(i)=0$, so $d_1(i)=0$ and then $d_0(i)=0$, contradicting $i \in M$. Hence the sets $M \cap G_\gamma$ are pairwise disjoint, and by (A3) nonempty, so $|M| \ge |E|$. Count pairs (i,γ) with $i \in M$, $\gamma \in E$, $i \notin G_\gamma$: by (A1) there are at most $|E|e$; each $i \in M$ lies in at most one G_γ, so there are at least $|M|(|E|-1) \ge |E|(|E|-1)$. Thus $|E|-1 \le e$. □

Only the field structure (to cancel $\gamma-\beta$) and (A1)–(A3) are used; c_0,c_1 need not be codewords. The bound is attained whenever $n \ge e+1$ and $q \ge e+1$: take challenges $z_1,\dots,z_{e+1}$, coordinates $x_1,\dots,x_{e+1}$, $d_1(x_j)=1$, $d_0(x_j)=-z_j$, $d=0$ elsewhere, and $G_{z_j} = \iota \setminus \{x_k : k \ne j\}$.

4 MCA from line decodability

Theorem 4. *Let $\delta < 1$, $b \geq \lfloor \delta n \rfloor + 2$, $b \geq 3$ and $n < q$. If C is (δ, a, b) line decodable, then $\varepsilon_{\mathrm{mca}}(C, \delta) \leq a/q$.*

Proof outline. We follow the proof of [1, Thm. 3.5] as formalized in ArkLib and change one step. Fix u, let B be the set of bad challenges and suppose $|B| > a$. For $\gamma \in B$ fix a witness set S_γ and codeword w_γ.

Nonzero code. Let J be the set of coordinates where some codeword is nonzero; $|J| \leq n < q$ gives a codeword d nonzero on all of J. The set $Z = \{i \notin J : u_0(i) = u_1(i) = 0\}$ has fewer than $n - e$ elements: otherwise $V(\gamma) = \gamma^2 d$ is close to the line for every γ, and line decodability would give at least $b \geq 3$ challenges with $\gamma^2 d = c_0 + \gamma c_1$, which happens for at most two. Define $V(\gamma) = w_\gamma$ on B; off B, if every codeword is close to the line, $V(\gamma) = t_\gamma d$ with t_γ chosen so that $V(\gamma)$ differs from the line on all of J (possible since $|J| < q$), and otherwise a codeword far from the line. V depends only on u and the chosen witnesses, and line decodability holds for every V. Since $|B| > a$, it yields c_0, c_1 and a set E of at least b challenges where $V(\gamma)$ is close to the line and equals $c_0 + \gamma c_1$. Every $\gamma \in E$ has a coordinate where the line equals the pencil but $u \neq (c_0, c_1)$: for $\gamma \in B$ inside S_γ, by the failure of the witness; for $\gamma \notin B$ in the agreement set of $t_\gamma d$ outside Z and J. This is where [1] uses an injection into ι to get $|E| \leq n$. Instead apply Lemma 3 with $G_\gamma = \{i : u_0(i) + \gamma u_1(i) = c_0(i) + \gamma c_1(i)\}$, which has at least $n - e$ elements because $V(\gamma) = c_0 + \gamma c_1$ is close to the line. So $|E| \leq e + 1 < b$, a contradiction.

Zero code. Lemma 3 with pencil $(0, 0)$ and $G_\gamma = S_\gamma$ gives $|B| \leq e + 1$; the synthetic construction of [1] produces $E \subseteq B$ with $|E| \geq b$.

All probabilities are exact counts over a uniform γ; no union bound is used. □

For $n \geq 2$ the hypotheses follow from $b = n + 1$, so Theorem 4 contains [1, Thm. 3.5] for lines. The condition $n < q$ replaces $n + 1 \leq q$, which $b = n + 1$ implies; it excludes Reed–Solomon codes on the whole field. Whether $b = e + 1$ suffices is open; the counting lemma is tight, but that does not settle the implication.

5 Boosting and folded Reed–Solomon codes

Lemma 5. *Under the hypotheses of the boosting lemma of [1] as formalized in ArkLib (`strong_LineDecodable_boost_of_lambda_le`: strong line decodability with parameters (a, b), $b > 1$, list size at most L at radius $\delta b/(b-1)$, $(L+1)^2 < q$), the code is strongly line decodable with parameters $((b' - 1)L + a,\ b')$ for every target b'.*

The existing proof is already parametric in the target: the at most L globally close pairs each align fewer than b' challenges, removing them costs at most $(b' - 1)L$, and the weak property applied to the rest gives a contradiction. The case $b' = n + 1$ is the existing statement.

Corollary 6. *With $m = \max(\lfloor \delta n \rfloor + 1, 2)$, strong line decodability with parameters $(a, 2t)$, list size at most t at radius $2t\delta/(2t-1)$, $n < q$ and $(t+1)^2 \leq mt + a$ give $\varepsilon_{\mathrm{mca}}(C, \delta) \leq (mt + a)/q$. The existing bound is $(nt + a)/q$.*

Corollary 7. *Under the hypotheses of [1, Cor. 4.10] as formalized in ArkLib (`frs_mcaError_le`: admissible folding with $s > 4t^2$, $sn < q$, a generator ω of $\mathbb{F}^\times$, $t > 0$), with $\rho = k/(sn)$ and $\delta = 1 - \rho - 2/t$,*

$$\varepsilon_{\mathrm{mca}}(C, \delta) \leq \frac{(\delta n + 2)t + 3t^3}{q},$$

and therefore $\varepsilon_{\mathrm{mca}}(C,\delta) \le \min\bigl(nt+3t^3,\ (\delta n+2)t+3t^3\bigr)/q$. *The second bound is smaller exactly when* $(1-\delta)n > 2$.

ρ	t	δ	n	ratio of bounds	bits
1/2	10	0.30	2^{20}	3.33	1.74
1/2	20	0.40	2^{20}	2.50	1.32
1/2	50	0.46	2^{20}	2.16	1.11
1/4	20	0.65	2^{20}	1.54	0.62
1/8	20	0.775	2^{20}	1.29	0.37

Table 1: Ratio $(nt+3t^3)/((\delta n+2)t+3t^3)$ of the existing and new FRS bounds. The gain is close to $1/\delta$ when $nt \gg t^3$. These are upper bounds on the MCA error, not measured soundness of a protocol.

6 Formalization

All statements are proved in Lean 4 (toolchain v4.35.0) against ArkLib commit `2203b01`. New declarations: `card_le_floor_add_one_of_pencil_mismatch` (Lemma 3), `IsLineDecodable.mcaError_le_of_floor_add_two` (Theorem 4), `strongLineDecodable_boost_of_lambda_le_target` (Lemma 5), `mcaError_le_of_strongLineDecodable_of_floor_add_two` (Corollary 6), `frs_mcaError_le_floor_add_two` and `frs_mcaError_le_min` (Corollary 7). The existing theorems keep their statements; in the final version `IsLineDecodable.mcaError_le` and the original boosting lemma are derived from the same parametrised proofs. `#print axioms` for every new declaration lists only `propext`, `Classical.choice` and `Quot.sound`; ArkLib's full validation script (`validate.sh --axioms`) passes. [Repository link and commit to be added at release.]

7 Related work

The threshold $\ell n+1$ appears in [1, Thm. 3.5] and in [3, Thm. 2.10]. The MCA bound of [2, Thm. 2.13] has the form $(\ell nL+a)/|K|$; we expect the same refinement to apply for $\ell = 1$ but have not checked it against its proof. Jeronimo, Liu and Rajpal [4] use a related agreement count for line correlated agreement. For degree-ℓ curves the counting argument gives $|E| \le \ell(e+1)$, since a mismatch coordinate is a root for at most ℓ challenges; we checked this numerically for small cases but did not formalize it.

Acknowledgements and use of AI tools

The counting lemma follows `exists_common_affine_set` of Proximity Prize submission `016274d1-9b09-4ce1-add3-4c632dbbc283` [5]. The formalization builds on ArkLib and its contributors. Claude (Anthropic) wrote most of the Lean code and the numerical checks; ChatGPT (OpenAI) reviewed the statements, quantifier structure and literature comparison. The author takes responsibility for the content and presentation of this note. The formalized theorems have been checked by the Lean kernel, while the exposition and formalization were developed with AI assistance as disclosed above.

References

[1] R. Goyal and V. Guruswami. Optimal proximity gaps for subspace-design codes and (random) Reed–Solomon codes. ECCC TR25-166 (2025, revised 2026); IACR ePrint 2025/2054.

[2] O. Alrabiah, R. Goyal and V. Guruswami. Explicit Reed–Solomon subspace designs: optimal list and line decoding. ECCC TR26-262; arXiv:2610.36215, October 2026.

[3] R. Goyal, V. Guruswami, Y. Sun and M. Wootters. Locality of curve-decoding and improved proximity gaps. arXiv:2607.08516, 2026; APPROX/RANDOM 2026, LIPIcs vol. 392, paper 61.

[4] F. G. Jeronimo, L. Liu and P. Rajpal. Optimal proximity gap for folded Reed–Solomon codes via subspace designs. arXiv:2601.10047, 2026.

[5] Proximity Prize submission `016274d1-9b09-4ce1-add3-4c632dbbc283` (68.15 bits), `Proxim_ityPrize/SubmissionLower` in `https://github.com/proximity-prize/proximity-prize`, commit `eb82d3b`, Apache-2.0.

[6] ArkLib contributors. ArkLib: a library for formally verified arguments of knowledge. `https://github.com/Verified-zkEVM/ArkLib`.